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REVISION

General Certificate of Education Advanced Level

MATHEMATICS SPECIMEN

6082/1

PAPER 1

**ZIMSEC 2026 SPECIMEN PAPER 2 SUGGESTED MARKING
GUIDE**

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Section A [40 Marks]
Answer **all** questions in this section.

Question 1

1(a) Simplify:

$$\frac{x^3\sqrt{x}}{3x^5 \times x^{-2}}$$

First write the cube root using indices:

$$\sqrt[3]{x} = x^{\frac{1}{3}}$$

So the numerator becomes:

$$x \times x^{\frac{1}{3}} = x^{1+\frac{1}{3}} = x^{\frac{4}{3}}$$

The denominator is:

$$3x^5 \times x^{-2} = 3x^{5-2} = 3x^3$$

Therefore,

$$\begin{aligned} \frac{x^{\frac{4}{3}}}{3x^3} &= \frac{1}{3} x^{\frac{4}{3}-3} \\ &= \frac{1}{3} x^{-\frac{5}{3}} \\ &= \boxed{\frac{1}{3x^{\frac{5}{3}}}} \end{aligned}$$

Mark allocation

Step	Mark
Correctly uses index laws for cube root and denominator	1
Final simplified answer	1
Total	2

1(b) y is inversely proportional to the cube root of $x + 1$.

So,

$$y \propto \frac{1}{\sqrt[3]{x+1}}$$
$$y = \frac{k}{\sqrt[3]{x+1}}$$

Given:

$$y = 2, x = 63$$

Substitute:

$$2 = \frac{k}{\sqrt[3]{63+1}}$$
$$2 = \frac{k}{\sqrt[3]{64}}$$
$$2 = \frac{k}{4}$$
$$k = 8$$

Therefore,

$$y = \frac{8}{\sqrt[3]{x+1}}$$

Mark allocation

Step	Mark
Correct proportional relationship	1
Correct final expression for y	1
Total	2

Question 2

Given:

$$U_{n+1} = 3 - U_n$$

2(a) When:

$$U_1 = 2$$

Then:

$$U_2 = 3 - U_1 = 3 - 2 = 1$$

$$U_3 = 3 - U_2 = 3 - 1 = 2$$

$$U_4 = 3 - U_3 = 3 - 2 = 1$$

So the sequence is:

$$2, 1, 2, 1, 2, 1, \dots$$

Therefore, the sequence is **periodic** and alternates between 2 and 1.

$$\boxed{2, 1, 2, 1, \dots}$$

Mark allocation

Step	Mark
Finds successive terms correctly	1
Describes alternating/periodic behaviour	1
Total	2

2(b) Given: $U_{n+1} = 3 - U_n$

Now,

$$U_{n+2} = 3 - U_{n+1}$$

Substitute:

$$U_{n+2} = 3 - (3 - U_n)$$

$$U_{n+2} = 3 - 3 + U_n$$

$$\boxed{U_{n+2} = U_n}$$

Now find U_{n+3} : $U_{n+3} = 3 - U_{n+2}$

Since: $U_{n+2} = U_n$

Then:

$$\boxed{U_{n+3} = 3 - U_n}$$

Mark allocation

Step	Mark
Shows $U_{n+2} = U_n$	1
Finds $U_{n+3} = 3 - U_n$	1
Total	2

Question 3

The repayments form an arithmetic sequence:

$$20, 22, 24, \dots, 114$$

First term: $a = 20$

Common difference: $d = 2$

Last term: $l = 114$

3(a) Use:

$$l = a + (n - 1)d$$

$$114 = 20 + (n - 1)(2)$$

$$114 - 20 = 2(n - 1)$$

$$94 = 2(n - 1)$$

$$47 = n - 1$$

$$n = 48$$

$$\boxed{48 \text{ months}}$$

Mark allocation

Step	Mark
Correct arithmetic sequence formula	1
Correct number of months	1
Total	2

3(b) The loan amount is the sum of the monthly repayments.

Use:

$$S_n = \frac{n}{2}(a + l)$$

$$S_{48} = \frac{48}{2}(20 + 114)$$

$$S_{48} = 24(134)$$

$$S_{48} = 3216$$

$$\boxed{\$3216}$$

Mark allocation

Step	Mark
Correct sum formula	1
Correct loan amount	1
Total	2

Question 4

The functions are:

$$f: x \mapsto 2 - \frac{1}{x}$$

$$g: x \mapsto \frac{1}{2x + 6}$$

4(a)(i) Find gf , meaning:

$$gf(x) = g(f(x))$$

Since:

$$f(x) = 2 - \frac{1}{x}$$

Then:

$$\begin{aligned} g(f(x)) &= \frac{1}{2f(x) + 6} \\ &= \frac{1}{2\left(2 - \frac{1}{x}\right) + 6} \end{aligned}$$

$$= \frac{1}{4 - \frac{2}{x} + 6}$$

$$= \frac{1}{10 - \frac{2}{x}}$$

Multiply top and bottom by x :

$$gf(x) = \frac{x}{10x - 2}$$

$gf(x) = \frac{x}{10x - 2}$

Mark allocation

Step	Mark
Correct substitution $g(f(x))$	1
Correct simplification	1
Total	2

4(a)(ii) Find $f^{-1}(x)$.

Let:

$$y = 2 - \frac{1}{x}$$

Rearrange:

$$y - 2 = -\frac{1}{x}$$

$$2 - y = \frac{1}{x}$$

$$x = \frac{1}{2 - y}$$

Now replace y by x :

$f^{-1}(x) = \frac{1}{2 - x}$

Mark allocation

Step	Mark
Correct rearrangement of $y = 2 - \frac{1}{x}$	1
Correct inverse function	1
Total	2

4(b) The graphs of a function and its inverse are reflections of each other in the line:

$$y = x$$

Therefore, the graphs of $g(x)$ and $g^{-1}(x)$ are reflections in the line:

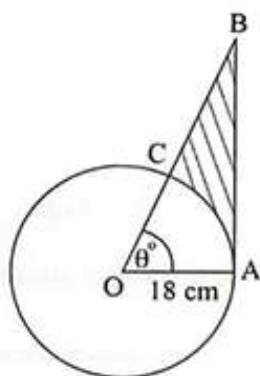
$$\boxed{y = x}$$

Mark allocation

Step	Mark
Correct geometrical relationship	1
Total	1

Question 5

The shaded region is bounded by the tangent AB , the line BC , and the arc AC .



Radius: $OA = OC = 18$

Angle: $AOC = \theta$

Since AB is tangent to the circle at A ,

$$OA \perp AB$$

So triangle OAB is right-angled at A .

In triangle OAB :

$$\begin{aligned}\tan\theta &= \frac{AB}{OA} \\ \tan\theta &= \frac{AB}{18} \\ AB &= 18\tan\theta\end{aligned}$$

Area of triangle OAB :

$$\begin{aligned}&\frac{1}{2} \times OA \times AB \\ &= \frac{1}{2} \times 18 \times 18\tan\theta \\ &= 162\tan\theta\end{aligned}$$

Area of sector AOC : $\frac{1}{2}r^2\theta$

$$\begin{aligned}&= \frac{1}{2}(18)^2\theta \\ &= 162\theta\end{aligned}$$

Therefore, shaded area: Area = area of triangle OAB – area of sector AOC
 $= 162\tan\theta - 162\theta$

$$\boxed{162(\tan\theta - \theta)}$$

where θ is in radians.

Mark allocation

Step	Mark
Recognises AB is tangent and $OA \perp AB$	1
Finds $AB = 18\tan\theta$	1
Finds area of triangle $OAB = 162\tan\theta$	1
Finds area of sector $AOC = 162\theta$	1
Correct shaded area $162(\tan\theta - \theta)$	1
Total	5

Question 6

Express $\frac{x^2+3x-13}{(x+2)(x-1)}$ in the form $A + \frac{B}{x+2} + \frac{C}{x-1}$

Since the numerator and denominator are both quadratic, first divide.

Denominator: $(x+2)(x-1) = x^2 + x - 2$

So:

$$\frac{x^2 + 3x - 13}{x^2 + x - 2}$$

Divide:

$$x^2 + 3x - 13 = (x^2 + x - 2) + (2x - 11)$$

Therefore:

$$\frac{x^2 + 3x - 13}{x^2 + x - 2} = 1 + \frac{2x - 11}{(x+2)(x-1)}$$

Now write:

$$\frac{2x - 11}{(x+2)(x-1)} = \frac{B}{x+2} + \frac{C}{x-1}$$

Multiply through by $(x+2)(x-1)$:

$$2x - 11 = B(x-1) + C(x+2)$$

Let $x = 1$:

$$2(1) - 11 = C(1+2)$$

$$-9 = 3C$$

$$C = -3$$

Let $x = -2$:

$$2(-2) - 11 = B(-2-1)$$

$$-15 = -3B$$

$$B = 5$$

Therefore:

$$\boxed{\frac{x^2 + 3x - 13}{(x+2)(x-1)} = 1 + \frac{5}{x+2} - \frac{3}{x-1}}$$

So:

$$A = 1, B = 5, C = -3$$

Mark allocation

Step	Mark
Expands denominator correctly	1
Performs division or separates constant term	1
Sets up partial fractions correctly	1
Finds $B = 5$	1
Finds $C = -3$ and final expression	1
Total	5

Question 7

Given: $P(x) = x^3 - ax^2 + x - b$

It is exactly divisible by $x - 1$, so: $P(1) = 0$

It leaves a remainder of -1 when divided by $x - 2$, so: $P(2) = -1$

7(a) Using $P(1) = 0$:

$$1^3 - a(1)^2 + 1 - b = 0$$

$$1 - a + 1 - b = 0$$

$$2 - a - b = 0$$

$$a + b = 2(1)$$

Using $P(2) = -1$:

$$2^3 - a(2)^2 + 2 - b = -1$$

$$8 - 4a + 2 - b = -1$$

$$10 - 4a - b = -1$$

$$4a + b = 11(2)$$

Now solve:

$$a + b = 2$$

$$4a + b = 11$$

Subtract:

$$3a = 9$$

$$a = 3$$

Substitute into:

$$a + b = 2$$

$$3 + b = 2$$

$$b = -1$$

$$\boxed{a = 3, b = -1}$$

Mark allocation

Step	Mark
Uses $P(1) = 0$ correctly	1
Uses $P(2) = -1$ correctly	1
Forms simultaneous equations	1
Solves for $a = 3, b = -1$	1
Total	4

7(b) Now:

$$P(x) = x^3 - 3x^2 + x - (-1)$$

$$P(x) = x^3 - 3x^2 + x + 1$$

Since $x - 1$ is a factor, divide by $x - 1$.

Using synthetic division (or long division):

$$x^3 - 3x^2 + x + 1 \div (x - 1)$$

Coefficients:

$$1, -3, 1, 1$$

Synthetic division by 1:

$$\begin{array}{r|rrrr}
 1 & 1 & -3 & 1 & 1 \\
 & & 1 & -2 & -1 \\
 \hline
 & 1 & -2 & -1 & 0
 \end{array}$$

So: $P(x) = (x - 1)(x^2 - 2x - 1)$

The quadratic does not factorise further using integer factors.

Therefore:

$$P(x) = (x - 1)(x^2 - 2x - 1)$$

Mark allocation

Step	Mark
Substitutes $a = 3$, $b = -1$ correctly	1
Correct complete factorization	1
Total	2

Question 8

Mr Moyo's house is: $A = (-5, 3)$

The well is: $P = (2, 1)$

The well is the midpoint of the road joining Mr Moyo's house to Mr Dube's house.

Let Mr Dube's house be: $D = (x, y)$

8(a) Using the midpoint formula:

$$\left(\frac{-5 + x}{2}, \frac{3 + y}{2}\right) = (2, 1)$$

So:

$$\begin{aligned}
 \frac{-5 + x}{2} &= 2 \\
 -5 + x &= 4 \\
 x &= 9
 \end{aligned}$$

And:

$$\frac{3+y}{2} = 1$$

$$3+y = 2$$

$$y = -1$$

Therefore, Mr Dube's house is: $(9, -1)$

Mark allocation

Step	Mark
Correct midpoint equation for x -coordinate	1
Correct midpoint equation for y -coordinate and final coordinates	1
Total	2

8(b) Distance from Mr Moyo's house to Mr Dube's house: $A = (-5, 3), D = (9, -1)$

Use distance formula: $AD = \sqrt{(9 - -5)^2 + (-1 - 3)^2}$

$$AD = \sqrt{14^2 + (-4)^2}$$

$$AD = \sqrt{196 + 16}$$

$$AD = \sqrt{212}$$

$$AD = 2\sqrt{53}$$

$$\boxed{2\sqrt{53}}$$

or approximately: $\boxed{14.6}$

Mark allocation

Step	Mark
Correct substitution into distance formula	1
Correct distance	1
Total	2

8(c) The line of points equidistant from Mr Moyo's house and Mr Dube's house is the **perpendicular bisector** of AD .

The midpoint is already given: $P = (2,1)$

Find the gradient of AD :

$$m_{AD} = \frac{-1 - 3}{9 - (-5)}$$

$$m_{AD} = \frac{-4}{14}$$

$$m_{AD} = -\frac{2}{7}$$

The gradient of the perpendicular bisector is the negative reciprocal:

$$m = \frac{7}{2}$$

Using: $y - y_1 = m(x - x_1)$ with point $(2,1)$:

$$y - 1 = \frac{7}{2}(x - 2)$$

Multiply by 2:

$$2y - 2 = 7x - 14$$

$$2y = 7x - 12$$

$$7x - 2y - 12 = 0$$

Therefore, the required equation is:

$$\boxed{7x - 2y - 12 = 0}$$

Mark allocation

Step	Mark
Recognises required line is perpendicular bisector	1
Finds perpendicular gradient $\frac{7}{2}$	1
Correct equation of the line	1
Total	3

Section B [80 Marks]

Answer **any five** questions from this section.

Question 9

9(a) Show that

$$\int_0^{\frac{\pi}{4}} \cos 2x \, dx = \frac{1}{2}$$

Integrate:

$$\int \cos 2x \, dx = \frac{1}{2} \sin 2x$$

Therefore,

$$\begin{aligned} \int_0^{\frac{\pi}{4}} \cos 2x \, dx &= \left[\frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{4}} \\ &= \frac{1}{2} \sin \left(2 \times \frac{\pi}{4} \right) - \frac{1}{2} \sin (0) \\ &= \frac{1}{2} \sin \frac{\pi}{2} - 0 \\ &= \frac{1}{2} (1) \\ &= \frac{1}{2} \end{aligned}$$

$$\boxed{\int_0^{\frac{\pi}{4}} \cos 2x \, dx = \frac{1}{2}}$$

Mark allocation

Step	Marks
Correct integration of cos 2x	1
Correct substitution of limits and final answer	1
Total	2

9(b) Find the exact value of

$$\int_0^{\frac{\pi}{4}} \sin^2 x \, dx$$

Use the identity: $\sin^2 x = \frac{1 - \cos 2x}{2}$

So,

$$\int_0^{\frac{\pi}{4}} \sin^2 x \, dx = \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2x}{2} \, dx = \frac{1}{2} \int_0^{\frac{\pi}{4}} 1 \, dx - \frac{1}{2} \int_0^{\frac{\pi}{4}} \cos 2x \, dx$$

From part (a),

$$\int_0^{\frac{\pi}{4}} \cos 2x \, dx = \frac{1}{2}$$

Also,

$$\begin{aligned} \frac{1}{2} \int_0^{\frac{\pi}{4}} 1 \, dx &= \frac{1}{2} [x]_0^{\frac{\pi}{4}} \\ &= \frac{1}{2} \left(\frac{\pi}{4} \right) = \frac{\pi}{8} \end{aligned}$$

Therefore,

$$\begin{aligned} \int_0^{\frac{\pi}{4}} \sin^2 x \, dx &= \frac{\pi}{8} - \frac{1}{2} \left(\frac{1}{2} \right) \\ &= \frac{\pi}{8} - \frac{1}{4} = \boxed{\frac{\pi - 2}{8}} \end{aligned}$$

Mark allocation

Step	Marks
Uses $\sin^2 x = \frac{1 - \cos 2x}{2}$	1
Correct integration setup	1
Correct use of result from part (a)	1
Correct exact answer	1
Total	4

9(c)(i) Use the trapezium rule with 2 strips to estimate:

$$\int_0^{\frac{\pi}{4}} \sec x \, dx$$

There are 2 strips, so:

$$h = \frac{\frac{\pi}{4} - 0}{2}$$
$$h = \frac{\pi}{8}$$

The x -values are:

$$0, \frac{\pi}{8}, \frac{\pi}{4}$$

Now find the corresponding y -values where:

$$y = \sec x$$

$$y_0 = \sec 0 = 1$$

$$y_1 = \sec \frac{\pi}{8}$$

$$y_2 = \sec \frac{\pi}{4} = \sqrt{2}$$

Using a calculator:

$$\sec \frac{\pi}{8} = \frac{1}{\cos \frac{\pi}{8}} \approx 1.0824$$

$$\sec \frac{\pi}{4} \approx 1.4142$$

Trapezium rule:

$$A \approx \frac{h}{2}(y_0 + 2y_1 + y_2)$$

$$A \approx \frac{\pi/8}{2}(1 + 2(1.0824) + 1.4142)$$

$$A \approx \frac{\pi}{16}(1 + 2.1648 + 1.4142)$$

$$A \approx \frac{\pi}{16}(4.5790)$$

$$A \approx 0.899$$

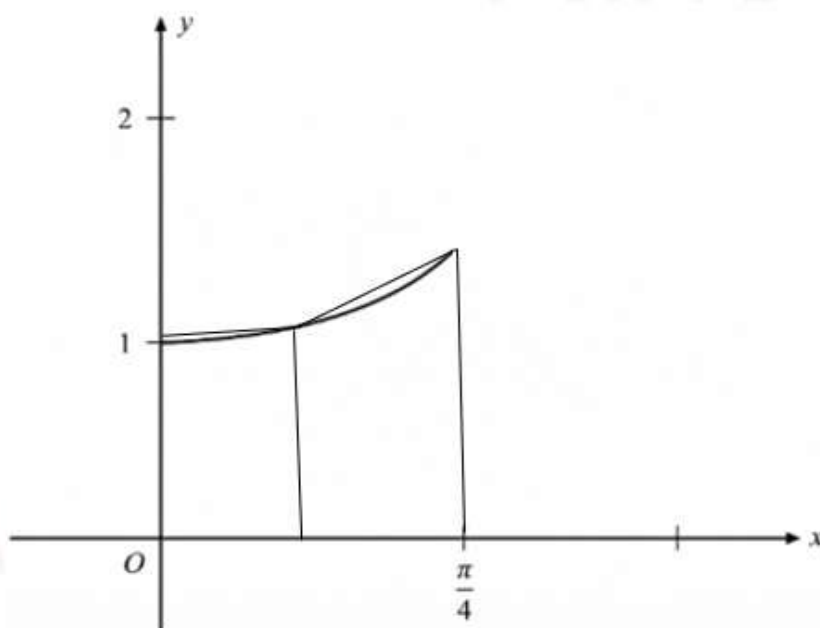
Correct to 2 significant figures:

$$\boxed{0.90}$$

Mark allocation

Step	Marks
Correct strip width $h = \frac{\pi}{8}$	1
Correct y -values	1
Correct trapezium rule formula	1
Correct estimate to 2 s.f.	1
Total	4

9(c)(ii) The graph of $y = \sec x$ on $0 \leq x \leq \frac{\pi}{4}$ is increasing and curves upwards.



Since the graph is **concave upwards**, the straight-line trapezia lie **above** the curve.

Therefore, the trapezium rule gives an **overestimate** of the area.

The trapezium rule overestimates the area.
--

Mark allocation

Step	Marks
Correctly identifies shape/concavity of $y = \sec x$	1
Explains trapezia lie above the curve	1
States overestimate	1
Total	3

9(d) Starting with: $x_0 = 2.5$

Use:

$$x_{r+1} = \frac{5}{x_r} + 1$$

Find x_1, x_2, x_3 .

First,

$$\begin{aligned}x_1 &= \frac{5}{x_0} + 1 \\x_1 &= \frac{5}{2.5} + 1 \\x_1 &= 2 + 1 \\x_1 &= 3\end{aligned}$$

Next,

$$\begin{aligned}x_2 &= \frac{5}{x_1} + 1 \\x_2 &= \frac{5}{3} + 1 \\x_2 &= \frac{5}{3} + \frac{3}{3} \\x_2 &= \frac{8}{3} \\x_2 &\approx 2.6667\end{aligned}$$

Next,

$$x_3 = \frac{5}{x_2} + 1$$

$$x_3 = \frac{5}{8/3} + 1$$

$$x_3 = \frac{15}{8} + 1$$

$$x_3 = \frac{23}{8}$$

$$x_3 = 2.875$$

Therefore,

$$x_1 = 3, x_2 = \frac{8}{3}, x_3 = \frac{23}{8}$$

Mark allocation

Step	Marks
Correct x_1	1
Correct x_2	1
Correct x_3	1
Total	3

Question 10

$$10(a) \ M = \begin{pmatrix} 2 & 1 & -1 \\ -1 & 0 & 3 \\ 2 & K & 4 \end{pmatrix}$$

10(a)(i) Matrix M has 3 rows and 3 columns.

$$3 \times 3$$

Mark allocation

Step	Marks
Correct order of matrix	1
Total	1

10(a)(ii) Find M^2 :

$$M^2 = M \times M$$

$$M^2 = \begin{pmatrix} 2 & 1 & -1 \\ -1 & 0 & 3 \\ 2 & K & 4 \end{pmatrix} \begin{pmatrix} 2 & 1 & -1 \\ -1 & 0 & 3 \\ 2 & K & 4 \end{pmatrix}$$

Now multiply row by column:

$$M^2 = \begin{pmatrix} 1 & 2-K & -3 \\ 4 & 3K-1 & 13 \\ 12-K & 4K+2 & 3K+14 \end{pmatrix}$$

$$M^2 = \begin{pmatrix} 1 & 2-K & -3 \\ 4 & 3K-1 & 13 \\ 12-K & 4K+2 & 3K+14 \end{pmatrix}$$

Mark allocation

Step	Marks
Correct matrix multiplication setup	1
Correct final M^2	1
Total	2

10(a)(iii) Given: $|M| = 20$

$$M = \begin{pmatrix} 2 & 1 & -1 \\ -1 & 0 & 3 \\ 2 & K & 4 \end{pmatrix}$$

Expand along the first row:

$$|M| = 2 \begin{vmatrix} 0 & 3 \\ K & 4 \end{vmatrix} - 1 \begin{vmatrix} -1 & 3 \\ 2 & 4 \end{vmatrix} + (-1) \begin{vmatrix} -1 & 0 \\ 2 & K \end{vmatrix}$$

Now calculate each determinant:

$$\begin{vmatrix} 0 & 3 \\ K & 4 \end{vmatrix} = 0(4) - 3K = -3K$$

$$\begin{vmatrix} -1 & 3 \\ 2 & 4 \end{vmatrix} = (-1)(4) - 3(2) = -4 - 6 = -10$$

$$\begin{vmatrix} -1 & 0 \\ 2 & K \end{vmatrix} = (-1)(K) - 0(2) = -K$$

Therefore,

$$\begin{aligned}
 |M| &= 2(-3K) - 1(-10) + (-1)(-K) \\
 |M| &= -6K + 10 + K \\
 |M| &= 10 - 5K
 \end{aligned}$$

Given:

$$\begin{aligned}
 10 - 5K &= 20 \\
 -5K &= 10 \\
 K &= -2 \\
 \boxed{K = -2}
 \end{aligned}$$

Mark allocation

Step	Marks
Correct determinant expansion	1
Correct simplified determinant 10 - 5K	1
Solves 10 - 5K = 20 correctly	1
Total	3

Question 10(b) Given:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix} A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -5 & -2 & 1 \end{pmatrix} \text{ and } B^{-1} = \begin{pmatrix} 1 & -4 & 10 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix}$$

10(b)(i) Find $(AB)^{-1}$

Use the rule $(AB)^{-1} = B^{-1}A^{-1}$

So,

$$(AB)^{-1} = \begin{pmatrix} 1 & -4 & 10 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -5 & -2 & 1 \end{pmatrix}$$

Now multiply:

First row:

$$\begin{aligned}
 1(1) + (-4)(1) + 10(-5) &= 1 - 4 - 50 = -53 \\
 1(0) + (-4)(1) + 10(-2) &= 0 - 4 - 20 = -24 \\
 1(0) + (-4)(0) + 10(1) &= 10
 \end{aligned}$$

Second row:

$$0(1) + 1(1) + (-3)(-5) = 1 + 15 = 16$$

$$0(0) + 1(1) + (-3)(-2) = 1 + 6 = 7$$

$$0(0) + 1(0) + (-3)(1) = -3$$

Third row:

$$0(1) + 0(1) + 1(-5) = -5$$

$$0(0) + 0(1) + 1(-2) = -2$$

$$0(0) + 0(0) + 1(1) = 1$$

Therefore,

$$(AB)^{-1} = \begin{pmatrix} -53 & -24 & 10 \\ 16 & 7 & -3 \\ -5 & -2 & 1 \end{pmatrix}$$

Mark allocation

Step	Marks
Uses $(AB)^{-1} = B^{-1}A^{-1}$	1
Correct matrix multiplication	1
Correct final matrix	1
Total	3

10(b)(ii) Solve:

$$x = 1$$

$$-x + y = 2$$

$$3x + 2y + z = 1$$

From the first equation: $x = 1$

Substitute into the second equation:

$$-1 + y = 2$$

$$y = 3$$

Substitute $x = 1$ and $y = 3$ into the third equation:

$$\begin{aligned}
 3(1) + 2(3) + z &= 1 \\
 3 + 6 + z &= 1 \\
 9 + z &= 1 \\
 z &= -8
 \end{aligned}$$

Therefore,

$$x = 1, y = 3, z = -8$$

Mark allocation

Step	Marks
Finds $x = 1$	1
Finds $y = 3$	1
Finds $z = -8$	1
Total	3

Question 10(c) Given $AP = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix}$, find matrix P .

Since:

$$AP = \text{given matrix}$$

Multiply both sides by A^{-1} :

$$\begin{aligned}
 P &= A^{-1}(AP) \\
 P &= \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -5 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix}
 \end{aligned}$$

Now multiply:

First row:

$$(1,0,0) \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix} = (1,1,1)$$

Second row:

$$\begin{aligned} (1,1,0) \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix} \\ = (1+2, 1+0, 1+1) \\ = (3,1,2) \end{aligned}$$

Third row:

$$\begin{aligned} (-5,-2,1) \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \end{pmatrix} \\ = (-5-4+0, -5+0+1, -5-2-2) \\ = (-9,-4,-9) \end{aligned}$$

Therefore,

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 3 & 1 & 2 \\ -9 & -4 & -9 \end{pmatrix}$$

Mark allocation

Step	Marks
Uses $P = A^{-1}(AP)$	1
Correct multiplication setup	1
Correct row calculations	1
Correct final matrix P	1
Total	

Question 11

11(a)(i) Prove that

$$\frac{1 - 2\cos^2 \theta}{\sin \theta \cos \theta} = \tan \theta - \cot \theta$$

Start with the left-hand side:

$$\frac{1 - 2\cos^2 \theta}{\sin \theta \cos \theta}$$

Since $1 = \sin^2 \theta + \cos^2 \theta$ then

$$\begin{aligned} 1 - 2\cos^2 \theta &= \sin^2 \theta + \cos^2 \theta - 2\cos^2 \theta \\ &= \sin^2 \theta - \cos^2 \theta \end{aligned}$$

So,

$$\frac{1 - 2\cos^2 \theta}{\sin \theta \cos \theta} = \frac{\sin^2 \theta - \cos^2 \theta}{\sin \theta \cos \theta}$$

Split the fraction:

$$\begin{aligned} &= \frac{\sin^2 \theta}{\sin \theta \cos \theta} - \frac{\cos^2 \theta}{\sin \theta \cos \theta} \\ &= \frac{\sin \theta}{\cos \theta} - \frac{\cos \theta}{\sin \theta} \\ &= \tan \theta - \cot \theta \end{aligned}$$

Hence proved.

$\frac{1 - 2\cos^2 \theta}{\sin \theta \cos \theta} = \tan \theta - \cot \theta$
--

Mark allocation

Step	Marks
Uses $1 = \sin^2 \theta + \cos^2 \theta$ correctly	1
Simplifies numerator to $\sin^2 \theta - \cos^2 \theta$	1
Splits fraction and obtains $\tan \theta - \cot \theta$	1
Total	3

11(a)(ii) Solve:

$$\frac{1 - 2\cos^2 \theta}{\sin \theta \cos \theta} = 2$$

Using part (i),

$$\tan \theta - \cot \theta = 2$$

Let $t = \tan\theta$

Since

$$\cot\theta = \frac{1}{\tan\theta} = \frac{1}{t}$$

we have

$$t - \frac{1}{t} = 2$$

Multiply by t :

$$\begin{aligned}t^2 - 1 &= 2t \\t^2 - 2t - 1 &= 0\end{aligned}$$

Solve using the quadratic formula:

$$\begin{aligned}t &= \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-1)}}{2} \\t &= \frac{2 \pm \sqrt{8}}{2} \\t &= \frac{2 \pm 2\sqrt{2}}{2} \\t &= 1 \pm \sqrt{2}\end{aligned}$$

So, $\tan\theta = 1 + \sqrt{2}$ or $\tan\theta = 1 - \sqrt{2}$

Now,

$$1 + \sqrt{2} = \tan 67.5^\circ$$

So,

$$\theta = 67.5^\circ, 247.5^\circ$$

Also,

$$1 - \sqrt{2} = -0.4142 \dots = \tan(-22.5^\circ)$$

So,

$$\theta = 157.5^\circ, 337.5^\circ$$

Therefore,

$$\boxed{\theta = 67.5^\circ, 157.5^\circ, 247.5^\circ, 337.5^\circ}$$

Mark allocation

Step	Marks
Uses result from part (i)	1
Forms equation $t - \frac{1}{t} = 2$	1
Solves quadratic $t^2 - 2t - 1 = 0$	1
Finds correct tangent values	1
Gives all four values of θ	1
Total	5

11(b)(i) Given:

$$x = 3 + 2\cos \theta$$
$$y = 4 + 2\sin \theta$$

Find $\frac{dy}{dx}$

Differentiate with respect to θ :

$$\frac{dx}{d\theta} = -2\sin\theta$$
$$\frac{dy}{d\theta} = 2\cos\theta$$

Therefore,

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$
$$= \frac{2\cos\theta}{-2\sin\theta}$$
$$= -\frac{\cos\theta}{\sin\theta}$$

$$\boxed{\frac{dy}{dx} = -\cot\theta}$$

Mark allocation

Step	Marks
Finds $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$	1
Correctly obtains $\frac{dy}{dx} = -\cot\theta$	1
Total	2

11(b)(ii) Show that the Cartesian equation is $(x - 3)^2 + (y - 4)^2 = 4$

From

$$\begin{aligned}x &= 3 + 2\cos\theta \\x - 3 &= 2\cos\theta\end{aligned}$$

Square both sides:

$$(x - 3)^2 = 4\cos^2\theta$$

From

$$\begin{aligned}y &= 4 + 2\sin\theta \\y - 4 &= 2\sin\theta\end{aligned}$$

Square both sides:

$$(y - 4)^2 = 4\sin^2\theta$$

Add:

$$\begin{aligned}(x - 3)^2 + (y - 4)^2 &= 4\cos^2\theta + 4\sin^2\theta \\&= 4(\cos^2\theta + \sin^2\theta)\end{aligned}$$

Since

$$\cos^2\theta + \sin^2\theta = 1$$

therefore,

$$\begin{aligned}(x - 3)^2 + (y - 4)^2 &= 4 \\ \boxed{(x - 3)^2 + (y - 4)^2 &= 4}\end{aligned}$$

Mark allocation

Step	Marks
Rearranges x and y correctly	1
Squares both equations	1
Adds equations	1
Uses $\sin^2 \theta + \cos^2 \theta = 1$ correctly	1
Total	4

11(b)(iii) $(x - 3)^2 + (y - 4)^2 = 4$

This is a circle with centre (3,4) and radius $\sqrt{4} = 2$

Circle with centre (3,4) and radius 2

Mark allocation

Step	Marks
Correctly describes centre and radius	1
Total	1

11(b)(iv) The line is $y = 2x$. The circle is $(x - 3)^2 + (y - 4)^2 = 4$

Substitute $y = 2x$:

$$(x - 3)^2 + (2x - 4)^2 = 4$$

Expand:

$$x^2 - 6x + 9 + 4x^2 - 16x + 16 = 4$$

$$5x^2 - 22x + 25 = 4$$

$$5x^2 - 22x + 21 = 0$$

Factorise:

$$5x^2 - 22x + 21 = 0$$

$$(5x - 7)(x - 3) = 0$$

So,

$$x = \frac{7}{5} \text{ or } x = 3$$

Since $y = 2x$,

When

$$x = \frac{7}{5}$$
$$y = \frac{14}{5}$$

When

$$x = 3$$
$$y = 6$$

Therefore, the coordinates are:

$$\left(\frac{7}{5}, \frac{14}{5}\right) \text{ and } (3, 6)$$

Mark allocation

Step	Marks
Substitutes $y = 2x$ into circle equation	1
Solves resulting quadratic	1
Finds correct coordinates	1
Total	3

Question 12

The position vectors are:

$$A = -2i + 3j - 6k = (-2, 3, -6)$$
$$B = -2i - j - 3k = (-2, -1, -3)$$
$$C = -i + 3j - 7k = (-1, 3, -7)$$

12(a) Find the distance between A and B .

$$\begin{aligned}\overrightarrow{AB} &= B - A \\ &= (-2, -1, -3) - (-2, 3, -6) \\ &= (0, -4, 3)\end{aligned}$$

Distance:

$$\begin{aligned}AB &= \sqrt{0^2 + (-4)^2 + 3^2} \\ &= \sqrt{0 + 16 + 9} \\ &= \sqrt{25} \\ \boxed{AB = 5}\end{aligned}$$

Mark allocation

Step	Marks
Finds vector $\overrightarrow{AB}$	1
Uses distance formula correctly	1
Correct distance	1
Total	3

12(b) Find the vector equation of the line from A to C .

$$\begin{aligned}A &= (-2, 3, -6) \\ C &= (-1, 3, -7)\end{aligned}$$

Direction vector:

$$\begin{aligned}\overrightarrow{AC} &= C - A \\ &= (-1, 3, -7) - (-2, 3, -6) \\ &= (1, 0, -1)\end{aligned}$$

Therefore, the vector equation is:

$$\mathbf{r} = \begin{pmatrix} -2 \\ 3 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

So, $\boxed{\mathbf{r} = \begin{pmatrix} -2 \\ 3 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}}$ where λ is a real parameter.

Mark allocation

Step	Marks
Identifies position vector of A	1
Finds direction vector $\overrightarrow{AC}$	1
Writes correct vector equation	1
Includes parameter	1
Total	4

12(c)

The two surveyed lines are AC and BC .

Direction vector of AC : $\overrightarrow{AC} = (1, 0, -1)$

Direction vector of BC : $\overrightarrow{BC} = C - B$
 $= (-1, 3, -7) - (-2, -1, -3)$
 $= (1, 4, -4)$

The angle θ between two vectors satisfies:

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

Now,

$$\mathbf{a} = (1, 0, -1), \mathbf{b} = (1, 4, -4)$$

Dot product:

$$\begin{aligned}\mathbf{a} \cdot \mathbf{b} &= 1(1) + 0(4) + (-1)(-4) \\ &= 1 + 0 + 4 = 5\end{aligned}$$

Magnitudes:

$$\begin{aligned}|\mathbf{a}| &= \sqrt{1^2 + 0^2 + (-1)^2} \\ &= \sqrt{2} \\ |\mathbf{b}| &= \sqrt{1^2 + 4^2 + (-4)^2} \\ &= \sqrt{1 + 16 + 16} \\ &= \sqrt{33}\end{aligned}$$

Therefore,

$$\cos\theta = \frac{5}{\sqrt{2}\sqrt{33}}$$

$$\cos\theta = \frac{5}{\sqrt{66}}$$

$$\theta = \cos^{-1}\left(\frac{5}{\sqrt{66}}\right)$$

$$\theta \approx 52.0^\circ$$

52.0°

Mark allocation

Step	Marks
Finds direction vectors of surveyed lines	1
Uses dot product formula	1
Calculates dot product and magnitudes correctly	1
Correct angle	1
Total	4

12(d) Find the equation of the plane containing A , B , and C .

Use two direction vectors in the plane:

$$\overrightarrow{AB} = (0, -4, 3)$$

$$\overrightarrow{AC} = (1, 0, -1)$$

A normal vector to the plane is:

$$\begin{aligned} & \overrightarrow{AB} \times \overrightarrow{AC} \\ &= \begin{vmatrix} i & j & k \\ 0 & -4 & 3 \\ 1 & 0 & -1 \end{vmatrix} \\ &= i[(-4)(-1) - 3(0)] - j[0(-1) - 3(1)] + k[0(0) - (-4)(1)] \\ &= 4i - (-3)j + 4k \\ &= (4, 3, 4) \end{aligned}$$

So the normal vector is $\mathbf{n} = (4, 3, 4)$

The plane has equation:

$$4x + 3y + 4z + d = 0$$

Use point $A = (-2, 3, -6)$:

$$4(-2) + 3(3) + 4(-6) + d = 0$$

$$-8 + 9 - 24 + d = 0$$

$$-23 + d = 0$$

$$d = 23$$

Therefore,

$$4x + 3y + 4z + 23 = 0$$

Mark allocation

Step	Marks
Finds two direction vectors in the plane	1
Uses cross product to find normal vector	2
Substitutes point into plane equation	1
Correct final plane equation	1
Total	5

Question 13

13(a) Method 1

Find:

$$\int e^{3x} \sin 4x \, dx$$

Use the standard result:

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C$$

Here,

$$a = 3, b = 4$$

So,

$$\begin{aligned}\int e^{3x} \sin 4x \, dx &= \frac{e^{3x}}{3^2 + 4^2} (3 \sin 4x - 4 \cos 4x) + C \\ &= \frac{e^{3x}}{25} (3 \sin 4x - 4 \cos 4x) + C\end{aligned}$$

$$\boxed{\int e^{3x} \sin 4x \, dx = \frac{e^{3x}}{25} (3 \sin 4x - 4 \cos 4x) + C}$$

Mark allocation

Step	Marks
Applies correct integration form or integration by parts method	2
Correct denominator $3^2 + 4^2 = 25$	1
Correct numerator $3 \sin 4x - 4 \cos 4x$	2
Includes constant of integration	1
Total	6

OR

Method 2: Integration by Parts

Find:

$$I = \int e^{3x} \sin 4x \, dx$$

Let

$$I = \int e^{3x} \sin 4x \, dx$$

Use integration by parts:

$$\int u \, dv = uv - \int v \, du$$

Take:

$$u = \sin 4x, dv = e^{3x} dx$$

Then:

$$du = 4 \cos 4x \, dx, v = \frac{1}{3} e^{3x}$$

So,

$$I = \frac{1}{3}e^{3x}\sin 4x - \frac{4}{3}\int e^{3x}\cos 4x \, dx$$

Let

$$J = \int e^{3x}\cos 4x \, dx$$

Then:

$$I = \frac{1}{3}e^{3x}\sin 4x - \frac{4}{3}J$$

Now find J :

$$J = \int e^{3x}\cos 4x \, dx$$

Use integration by parts again.

Take:

$$u = \cos 4x, dv = e^{3x} dx$$

Then:

$$du = -4\sin 4x \, dx, v = \frac{1}{3}e^{3x}$$

So,

$$\begin{aligned} J &= \frac{1}{3}e^{3x}\cos 4x - \int \frac{1}{3}e^{3x}(-4\sin 4x) \, dx \\ J &= \frac{1}{3}e^{3x}\cos 4x + \frac{4}{3}\int e^{3x}\sin 4x \, dx \end{aligned}$$

But

$$I = \int e^{3x}\sin 4x \, dx$$

Therefore,

$$J = \frac{1}{3}e^{3x}\cos 4x + \frac{4}{3}I$$

Substitute into

$$\begin{aligned} I &= \frac{1}{3}e^{3x}\sin 4x - \frac{4}{3}J \\ I &= \frac{1}{3}e^{3x}\sin 4x - \frac{4}{3}\left(\frac{1}{3}e^{3x}\cos 4x + \frac{4}{3}I\right) \\ I &= \frac{1}{3}e^{3x}\sin 4x - \frac{4}{9}e^{3x}\cos 4x - \frac{16}{9}I \end{aligned}$$

Bring I -terms together:

$$\begin{aligned} I + \frac{16}{9}I &= \frac{1}{3}e^{3x}\sin 4x - \frac{4}{9}e^{3x}\cos 4x \\ \frac{25}{9}I &= \frac{1}{3}e^{3x}\sin 4x - \frac{4}{9}e^{3x}\cos 4x \end{aligned}$$

Multiply by $\frac{9}{25}$:

$$\begin{aligned} I &= \frac{9}{25}\left(\frac{1}{3}e^{3x}\sin 4x - \frac{4}{9}e^{3x}\cos 4x\right) \\ I &= \frac{3}{25}e^{3x}\sin 4x - \frac{4}{25}e^{3x}\cos 4x \end{aligned}$$

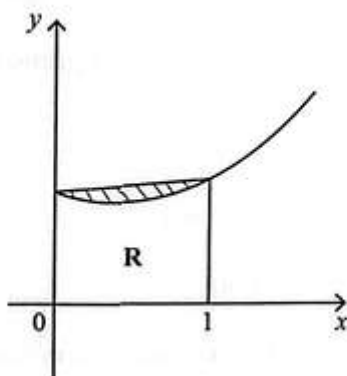
$$\int e^{3x}\sin 4x \, dx = \frac{e^{3x}}{25}(3\sin 4x - 4\cos 4x) + C$$

Mark allocation

Step	Marks
Defines $I = \int e^{3x}\sin 4x \, dx$ and applies integration by parts first time	2
Defines $J = \int e^{3x}\cos 4x \, dx$	1
Applies integration by parts to J correctly	1
Substitutes J back into I	1
Solves for I and gives final answer with $+C$	1
Total	6

13(b)(i)

The curve is $y = 1 + x^2$. Region R is bounded by the curve, the axes, and the line $x = 1$.



When the region is rotated about the x -axis, volume is:

$$V = \pi \int_0^1 y^2 dx$$

Since $y = 1 + x^2$, then

$$V = \pi \int_0^1 (1 + x^2)^2 dx$$

Expand $(1 + x^2)^2 = 1 + 2x^2 + x^4$

So,

$$V = \pi \int_0^1 (1 + 2x^2 + x^4) dx$$

$$V = \pi \left[x + \frac{2x^3}{3} + \frac{x^5}{5} \right]_0^1$$

$$V = \pi \left(1 + \frac{2}{3} + \frac{1}{5} \right)$$

LCM of 3 and 5 is 15:

$$\begin{aligned} 1 + \frac{2}{3} + \frac{1}{5} &= \frac{15}{15} + \frac{10}{15} + \frac{3}{15} \\ &= \frac{28}{15} \end{aligned}$$

Therefore,

$$V = \frac{28\pi}{15} \text{ cubic units}$$

Mark allocation

Step	Marks
Uses volume formula $V = \pi \int y^2 dx$	1
Substitutes $y = 1 + x^2$	1
Correct expansion and integration	1
Correct final answer $\frac{28\pi}{15}$	1
Total	4

13(b)(ii) Use the trapezium rule with 4 trapezia to estimate the area of R .

Area:

$$A = \int_0^1 (1 + x^2) dx$$

Number of trapezia: $n = 4$

Step size:

$$h = \frac{1 - 0}{4}$$
$$h = 0.25$$

The x -values are: 0, 0.25, 0.50, 0.75, 1

Now calculate $y = 1 + x^2$.

$$y_0 = 1 + 0^2 = 1$$
$$y_1 = 1 + (0.25)^2 = 1.0625$$
$$y_2 = 1 + (0.50)^2 = 1.25$$
$$y_3 = 1 + (0.75)^2 = 1.5625$$
$$y_4 = 1 + 1^2 = 2$$

Using the trapezium rule:

$$A \approx \frac{h}{2} [y_0 + 2(y_1 + y_2 + y_3) + y_4]$$
$$A \approx \frac{0.25}{2} [1 + 2(1.0625 + 1.25 + 1.5625) + 2]$$
$$A \approx 0.125 [1 + 2(3.875) + 2]$$
$$A \approx 0.125 [1 + 7.75 + 2]$$

$$A \approx 0.125(10.75)$$

$$A \approx 1.34375$$

Correct to 3 significant figures:

1.34 square units

Mark allocation

Step	Marks
Correct step size $h = 0.25$	1
Correct table of y -values	1
Correct trapezium rule substitution	1
Correct answer to 3 s.f.	1
Total	4

13(b)(iii) The curve $y = 1 + x^2$ concave upwards. For a concave upward curve, the trapezia lie above the curve. Therefore, the trapezium rule gives an overestimate.

The trapezium rule overestimates the area.

Mark allocation

Step	Marks
States curve is concave upwards	1
Correctly concludes overestimate	1
Total	2

Question 14

The rate of change of temperature is proportional to the temperature X .

So,

$$\frac{dX}{dt} \propto -X$$

The negative sign is used because the metal is cooling.

Thus,

$$\frac{dX}{dt} = -kX$$

where $k > 0$.

14(a) Show that:

$$X = 82e^{-kt}$$

Separate variables:

$$\frac{dX}{X} = -k dt$$

Integrate both sides:

$$\int \frac{1}{X} dX = \int -k dt$$
$$\ln X = -kt + C$$

Write in exponential form:

$$X = e^{-kt+C}$$
$$X = e^C e^{-kt}$$

Let $e^C = A$

So,

$$X = Ae^{-kt}$$

Initially $X = 82$ when $t = 0$

Therefore,

$$82 = Ae^0$$
$$A = 82$$

Hence,

$$\boxed{X = 82e^{-kt}}$$

Mark allocation

Step	Marks
Forms differential equation $\frac{dX}{dt} = -kX$	1
Separates variables	1
Integrates correctly	1
Writes exponential form	1
Uses initial condition $X = 82$ when $t = 0$	1
Obtains $X = 82e^{-kt}$	1
Total	6

14(b) Given:

$$X = 42 \text{ when } t = 10$$

Using:

$$\begin{aligned} X &= 82e^{-kt} \\ 42 &= 82e^{-10k} \\ e^{-10k} &= \frac{42}{82} \end{aligned}$$

Take natural logarithms:

$$\begin{aligned} -10k &= \ln\left(\frac{42}{82}\right) \\ k &= -\frac{1}{10} \ln\left(\frac{42}{82}\right) \\ k &\approx 0.0669 \end{aligned}$$

Now find the time when:

$$\begin{aligned} X &= 37 \\ 37 &= 82e^{-kt} \\ e^{-kt} &= \frac{37}{82} \\ -kt &= \ln\left(\frac{37}{82}\right) \\ t &= -\frac{\ln\left(\frac{37}{82}\right)}{k} \\ t &\approx 11.895 \text{ minutes} \end{aligned}$$

This is the total time from removal from the oven.

The question asks how much longer after 10 minutes:

$$11.895 - 10 = 1.895$$

1.89 minutes longer

or approximately

1 minute 54 seconds

Mark allocation

Step	Marks
Uses $42 = 82e^{-10k}$	1
Finds k correctly	1
Finds total time for $X = 37$	1
Subtracts 10 minutes to find extra time	1
Total	4

14(c)(i) Estimate the temperature after 3.2 minutes.

Use $X = 82e^{-kt}$, where $k \approx 0.0669$ and $t = 3.2$.

$$X = 82e^{-0.0669(3.2)}$$

$$X = 82e^{-0.2141}$$

$$X \approx 66.2$$

66.2°C

Mark allocation

Step	Marks
Substitutes $t = 3.2$ into cooling model	1
Correct temperature estimate	1
Total	2

14(c)(ii) The ironsmith heats the metal so that it becomes softer and easier to shape, bend, or forge into hoes and axes.

Heating softens the metal, making it easier to shape.

Mark allocation

Step	Marks
Gives sensible reason linked to softening/shaping metal	1
Total	1

14(d) Given $\ln y = xy, y > 0$. Show that:

$$\frac{dy}{dx} = \frac{y^2}{1 - xy}$$

Differentiate implicitly:

$$\begin{aligned}\frac{d}{dx}(\ln y) &= \frac{d}{dx}(xy) \\ \frac{1}{y} \frac{dy}{dx} &= x \frac{dy}{dx} + y\end{aligned}$$

Let:

$$\frac{dy}{dx} = y'$$

Then:

$$\frac{y'}{y} = xy' + y$$

Multiply by y :

$$y' = xyy' + y^2$$

Bring terms involving y' together:

$$y' - xyy' = y^2$$

Factorise:

$$y'(1 - xy) = y^2$$

Therefore,

$$\boxed{\frac{dy}{dx} = \frac{y^2}{1 - xy}}$$

Mark allocation

Step	Marks
Differentiates ln y correctly	1
Differentiates xy using product rule	1
Rearranges to obtain $\frac{dy}{dx} = \frac{y^2}{1 - xy}$	1
Total	3

Question 15

15(a) Given $\sin x = x^2 e^y$. Show that:

$$\frac{dy}{dx} = \frac{\cos x}{x^2 e^y} - \frac{2}{x}$$

Differentiate both sides:

$$\begin{aligned}\frac{d}{dx}(\sin x) &= \frac{d}{dx}(x^2 e^y) \\ \cos x &= 2xe^y + x^2 e^y \frac{dy}{dx}\end{aligned}$$

Now isolate $\frac{dy}{dx}$:

$$x^2 e^y \frac{dy}{dx} = \cos x - 2xe^y$$

Divide by $x^2 e^y$:

$$\frac{dy}{dx} = \frac{\cos x}{x^2 e^y} - \frac{2xe^y}{x^2 e^y}$$

$$\frac{dy}{dx} = \frac{\cos x}{x^2 e^y} - \frac{2}{x}$$

$$\boxed{\frac{dy}{dx} = \frac{\cos x}{x^2 e^y} - \frac{2}{x}}$$

Mark allocation

Step	Marks
Differentiates left side correctly	1
Uses product rule/chain rule correctly and rearranges	1
Total	2

15(b) Given:

$$x = \frac{t^2}{1-t}$$
$$y = \frac{t}{1+t^2}$$

Find $\frac{dy}{dx}$.

First find:

$$\frac{dx}{dt}$$

Using the quotient rule:

$$x = \frac{t^2}{1-t}$$
$$\frac{dx}{dt} = \frac{(1-t)(2t) - t^2(-1)}{(1-t)^2}$$
$$= \frac{2t - 2t^2 + t^2}{(1-t)^2}$$
$$= \frac{2t - t^2}{(1-t)^2}$$
$$= \frac{t(2-t)}{(1-t)^2}$$

Now find:

$$\frac{dy}{dt}$$

$$y = \frac{t}{1+t^2}$$

$$\frac{dy}{dt} = \frac{(1+t^2)(1) - t(2t)}{(1+t^2)^2}$$

$$= \frac{1+t^2-2t^2}{(1+t^2)^2}$$

$$= \frac{1-t^2}{(1+t^2)^2}$$

Therefore,

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$= \frac{1-t^2}{(1+t^2)^2} \div \frac{t(2-t)}{(1-t)^2}$$

$$= \frac{1-t^2}{(1+t^2)^2} \times \frac{(1-t)^2}{t(2-t)}$$

Since $1-t^2 = (1-t)(1+t)$ then

$$\frac{dy}{dx} = \frac{(1-t)^2(1-t)(1+t)}{t(2-t)(1+t^2)^2}$$

$$\boxed{\frac{dy}{dx} = \frac{(1-t)^3(1+t)}{t(2-t)(1+t^2)^2}}$$

Mark allocation

Step	Marks
Correctly finds $\frac{dx}{dt}$	1
Correctly finds $\frac{dy}{dt}$	1
Correctly forms and simplifies $\frac{dy}{dx}$	1
Total	3

15(c) Given $y = 4x^3 - x^2 - 4x + 10$

Find turning points and determine their nature.

Differentiate:

$$\frac{dy}{dx} = 12x^2 - 2x - 4$$

At turning points:

$$\frac{dy}{dx} = 0$$

$$\text{So } 12x^2 - 2x - 4 = 0$$

Divide by 2:

$$6x^2 - x - 2 = 0$$

Factorise:

$$\begin{aligned} 6x^2 - x - 2 &= 0 \\ (3x - 2)(2x + 1) &= 0 \end{aligned}$$

So,

$$\begin{aligned} 3x - 2 &= 0 \\ x &= \frac{2}{3} \end{aligned}$$

or

$$\begin{aligned} 2x + 1 &= 0 \\ x &= -\frac{1}{2} \end{aligned}$$

Now find the corresponding y -values.

When

$$\begin{aligned} x &= \frac{2}{3} \\ y &= 4\left(\frac{2}{3}\right)^3 - \left(\frac{2}{3}\right)^2 - 4\left(\frac{2}{3}\right) + 10 \\ y &= 4\left(\frac{8}{27}\right) - \frac{4}{9} - \frac{8}{3} + 10 \\ y &= \frac{32}{27} - \frac{12}{27} - \frac{72}{27} + \frac{270}{27} \end{aligned}$$

$$y = \frac{218}{27}$$

So one turning point is:

$$\left(\frac{2}{3}, \frac{218}{27}\right)$$

When

$$x = -\frac{1}{2}$$

$$y = 4\left(-\frac{1}{2}\right)^3 - \left(-\frac{1}{2}\right)^2 - 4\left(-\frac{1}{2}\right) + 10$$

$$y = 4\left(-\frac{1}{8}\right) - \frac{1}{4} + 2 + 10$$

$$y = -\frac{1}{2} - \frac{1}{4} + 12$$

$$y = \frac{45}{4}$$

So the other turning point is $\left(-\frac{1}{2}, \frac{45}{4}\right)$

Now determine the nature.

Second derivative:

$$\frac{d^2y}{dx^2} = 24x - 2$$

At

$$x = -\frac{1}{2}$$

$$\frac{d^2y}{dx^2} = 24\left(-\frac{1}{2}\right) - 2$$

$$= -12 - 2 = -14 < 0$$

So $\left(-\frac{1}{2}, \frac{45}{4}\right)$ is a maximum point.

At

$$x = \frac{2}{3}$$

$$\frac{d^2y}{dx^2} = 24\left(\frac{2}{3}\right) - 2$$

$$= 16 - 2 = 14 > 0$$

So,

$$\left(\frac{2}{3}, \frac{218}{27}\right)$$

is a minimum point.

$$\left(-\frac{1}{2}, \frac{45}{4}\right) \text{ is a maximum point}$$

$$\left(\frac{2}{3}, \frac{218}{27}\right) \text{ is a minimum point}$$

Mark allocation

Step	Marks
Correctly differentiates y	1
Sets $\frac{dy}{dx} = 0$	1
Solves for $x = -\frac{1}{2}, \frac{2}{3}$	1
Finds both y -coordinates	1
Uses second derivative or sign test	1
Correctly identifies maximum and minimum	1
Total	6

15(d) Given $f(x) = e^x \sin x$.

Find the first 3 terms in the Maclaurin series.

Use the expansions:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

$$\sin x = x - \frac{x^3}{6} + \dots$$

Therefore,

$$e^x \sin x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots\right) \left(x - \frac{x^3}{6} + \dots\right)$$

Multiply terms up to x^3 :

$$\begin{aligned} 1(x) &= x \\ x(x) &= x^2 \\ \frac{x^2}{2}(x) &= \frac{x^3}{2} \\ 1\left(-\frac{x^3}{6}\right) &= -\frac{x^3}{6} \end{aligned}$$

So,

$$\begin{aligned} e^x \sin x &= x + x^2 + \frac{x^3}{2} - \frac{x^3}{6} + \dots \\ &= x + x^2 + \frac{3x^3 - x^3}{6} + \dots \\ &= x + x^2 + \frac{2x^3}{6} + \dots \\ &= x + x^2 + \frac{x^3}{3} + \dots \end{aligned}$$

Therefore, the first 3 terms are:

$$e^x \sin x = x + x^2 + \frac{x^3}{3} + \dots$$

Mark allocation

Step	Marks
Writes correct series for e^x	1
Writes correct series for $\sin x$	1
Multiplies series correctly	2
Gives first 3 terms correctly	1
Total	5

OR

$$f(x) = e^x \sin x.$$

The Maclaurin series is given by:

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots$$

Find $f(0)$

$$\begin{aligned} f(0) &= e^0 \sin 0 \\ &= 1(0) \\ &= 0 \end{aligned}$$

Find the first derivative

$$f'(x) = \frac{d}{dx}(e^x \sin x)$$

Using the product rule:

$$\begin{aligned} f'(x) &= e^x \sin x + e^x \cos x \\ f'(x) &= e^x (\sin x + \cos x) \end{aligned}$$

At $x = 0$:

$$\begin{aligned} f'(0) &= e^0 (\sin 0 + \cos 0) \\ &= 1(0 + 1) \\ &= 1 \end{aligned}$$

Find the second derivative

$$f''(x) = \frac{d}{dx}[e^x (\sin x + \cos x)]$$

Using the product rule again:

$$\begin{aligned} f''(x) &= e^x (\sin x + \cos x) + e^x (\cos x - \sin x) \\ f''(x) &= e^x (2\cos x) \\ f''(x) &= 2e^x \cos x \end{aligned}$$

At $x = 0$:

$$\begin{aligned} f''(0) &= 2e^0 \cos 0 \\ &= 2(1)(1) \\ &= 2 \end{aligned}$$

Find the third derivative

$$f'''(x) = \frac{d}{dx}(2e^x \cos x)$$

Using the product rule:

$$f'''(x) = 2e^x \cos x - 2e^x \sin x$$

$$f'''(x) = 2e^x (\cos x - \sin x)$$

At $x = 0$:

$$f'''(0) = 2e^0 (\cos 0 - \sin 0)$$

$$= 2(1 - 0)$$

$$= 2$$

Substitute into the Maclaurin formula

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$= 0 + x(1) + \frac{x^2}{2} (2) + \frac{x^3}{6} (2) + \dots$$

$$= x + x^2 + \frac{x^3}{3} + \dots$$

Therefore, the first 3 terms are:

$$e^x \sin x = x + x^2 + \frac{x^3}{3} + \dots$$

Mark Allocation

Step	Marks
Correct Maclaurin formula	1
Finding $f(0)$	1
Correct first derivative and $f'(0)$	2
Correct second derivative and $f''(0)$	2
Correct third derivative and $f'''(0)$	2
Substitution into Maclaurin series	1
Final simplified answer	1
Total	10



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1 Corinthians 9 vs. 25

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